

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Chandubhai S Patel Institute of Technology**3rd Semester B.Tech.(EC/ME/CL/EE)

University Exam Nov-Dec 2011

MA-202 Engineering Mathematics-III

Date: 28/11/'11

Time: 01:30 pm to 04:30 pm

Maximum Marks: 70

Instruction:

1. This question paper contains two sections **Section-I** and **Section-II**.
2. **Section-I** carries 35 and **Section-II** carries 35 marks.
3. Section wise separate answer book to be used.
4. Figure to the right indicates the distribution of full marks for the respective question.

Section-I

- Q.1** (A) Find the Fourier series to represent e^{ax} in the interval $-\pi \leq x \leq \pi$. [3]
- (B) State convolution theorem and using it evaluate $L^{-1}\left\{\frac{a}{s^2(s^2+a^2)}\right\}$. [3]
- (C) Define the half-range sine series for the function $f(x)$ on $(0, l)$ and find the half-range cosine series for the function $f(x) = (x-1)^2$ in the interval $0 < x < 1$.
Hence show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$. [3]
- Q.2** (A) Find the Laplace transform of the function $te^{-4t}\sin 3t$. [4]
- (B) Solve $y'' + 4y' + 3y = e^{-t}$, $y(0) = 0$, $y'(0) = 1$ by using Laplace transform. [4]
- (C) Find Laplace transform of function given by $f(t) = \frac{kt}{p}$, if $0 < t < p$, $f(t+p) = f(t)$. [4]

OR

- Q.2** (A) Find the Laplace transform of the function $f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 2-t, & 1 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$ [4]
- (B) Solve $y'' - 3y' + 2y = 4t + e^{3t}$, $y(0) = 1$, $y'(0) = -1$ by using Laplace transform. [4]
- (C) Find inverse Laplace transform of $\frac{s^3}{s^4 - 81}$. [4]

- Q.3** (A) Solve $x^3 - 27x + 54 = 0$ by Cardon's Method. [5]
- (B) Find Fourier series to represent x^2 in the interval $-l \leq x \leq l$ and show that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$. [5]
- (C) Find the roots of the equation $x^4 + x^3 + x^2 + 3x - 6 = 0$. [4]

OR

- Q.3** (A) Solve $x^4 - 12x^3 + 41x^2 - 18x - 72 = 0$ by Ferrari's Method. [5]
- (B) Graph the following functions and find their corresponding Fourier series for $p = 2l = 4$, $f(x) = \begin{cases} 8, & 0 < x < 2; \\ -8, & 2 < x < 4. \end{cases}$ [5]
- (C) For which values of x , the expression $x^3 + 2x^2 - 5x - 6$ vanishes. [4]

Section-II

- Q.4 (A) (i) State Stock's theorem. [2]
 (ii) Define Laplace equation of two dimension and one dimensional Heat equation. [2]
 (B) An electrical circuit consists of an inductance of 0.1 henry, a resistance of 20 ohms and a condenser of capacitance 25 micro-farads. Find the charge q and the current i at any time t , given that at $t = 0$, $q = 0.05$ coulomb, $i = \frac{dq}{dt} = 0$ when $t = 0$ if we have $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0$. [4]

OR

- (B) A tightly stretched string of length l with fixed ends is initially in equilibrium position. It is set vibrating by given each point a velocity $v_0 \sin^3\left(\frac{\pi x}{l}\right)$. Find the displacement $y(x, t)$. [4]
 Q.5 (A) A vector field is given by $\vec{F} = 2x^2y\hat{i} + 3xy\hat{j}$. Evaluate the line integration $\int_C \vec{F} \cdot d\vec{r}$ from $(0,0)$ to $(1,4)$ along the curves $y = 4x^2$ and $y = 4x$. [4]
 (B) Find the unit normal vector to the surface $\nabla \cdot \vec{F}$ where $\vec{F} = (x^3 + yz)\hat{i} + (xz + y^4)\hat{j} + (z^5 + x^2y^3)\hat{k}$ at the point $(1, -1, 2)$. [4]
 (C) Apply the divergence theorem to compute $\iiint_S \vec{u} \cdot \hat{n} ds$, where S is the surface of the cylinder $x^2 + y^2 = 9$ bounded by the planes $z = 0, z = b$ where $\vec{u} = \hat{i}x - \hat{j}y + \hat{k}z$. [4]

OR

- Q.5 (A) Evaluate $\int \vec{A} \times d\vec{r}$ from $t = 0$ to $t = 1$ where $\vec{A} = xy\hat{i} - z\hat{j} + x^2\hat{k}$, $x = t^2, y = 2t, z = t^3$, and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. [4]
 (B) If $u = x^2yz, v = xy - 3z^2$, then find (i) $\text{grad}(\nabla u \cdot \nabla v)$ (ii) $\text{div}(\nabla u \times \nabla v)$. [4]
 (C) Use Green's theorem in the planes to evaluates $\int_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region defined by $x = y^2, y = x^2$. [4]

Q.[6] Attempt any **Three**. [15]

- (A) Solve the partial differential equation $u_t = c^2 u_{xx}$ for the conduction of heat along a rod of length l without radiation subject to the constrain $u(0, t) = u(l, t) = 0$ and $u(x, 0) = f(x)$.
 (B) Evaluate $\iint_S (\nabla \times \vec{F}) \cdot \hat{n} ds$ where $\vec{F} = x^2\hat{i} - xy\hat{j}$ and S is the square in plane $z = 0$ and bounded by the lines $x = 0, y = 0, x = a, y = a$.
 (C) Find the directional derivative of $\vec{V}^2$, where $\vec{V} = xy^2\hat{i} + zy^2\hat{j} + xz^2\hat{k}$, at the point $(2,0,3)$ in the direction of the out word normal to the sphere $x^2 + y^2 + z^2 = 14$ at the point $(3,2,1)$.
 (D) Show that $\vec{F} = (y^2 - z^2 + 3yx - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is conservative and find (a) scalar potential (b) the work done by $\vec{F}$ in moving a particle from $A(1,0,1)$ to $B(2,1,3)$.
 (E) (i) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $x^2 + y^2 - z = 3$ at the point $(2, -1, 2)$.
 (ii) If the directional derivative of $\phi = axy + byz + czx$ at $(1,1,1)$ has maximum magnitude 4 in a direction parallel to x-axis, find the value of a, b, c .